

Variance-Swap Targets

Log-contract replication, the LQD closed form, and the source-PDE method

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Abstract

A variance-swap quote is one number per smile node — the market’s fair var-swap volatility for an (underlying, T) — and it pins a global property of the smile that scattered option quotes constrain only weakly: the total risk-neutral variance, dominated by the wings. This note documents how the Vol-Fitter turns a var-swap quote into a soft calibration target shared by every model. The fair var-swap total variance is the model-free OTM log-contract replication; the penalty pulls the model’s *own* fair var-swap toward the quote, in vol units, weighted as a chosen fraction of the node’s option weight. Computing the model’s var-swap cheaply is the crux: the generic replication re-solves implied variance on a grid every Jacobian column and would make a single fit take minutes, so LQD uses its *exact closed form* ($-2\mathbb{E}[X]$) and the local-vol surface uses a *source PDE* ($I(T) = g(0, 1)$) with analytic sensitivities. A fresh run confirms the LQD closed form agrees with replication to 1.53 vol bps, the replication converges by a half-width of ~ 4 , and a var-swap quote 25.97% pulls a smile whose natural var-swap is 24.47% up to 25.91% while keeping the option fit.

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1 What a variance swap pins

A variance swap pays the realized variance of the underlying to T against a fixed strike. Under a diffusion the fair strike equals the risk-neutral expected total variance, which the model-free Carr–Madan log-contract replication expresses as a portfolio of OTM options:

$$w_{\text{vs}}(T) = \mathbb{E}^T \left[\int_0^T \sigma_t^2 dt \right] = 2 \int_0^\infty \frac{P(T, K)}{K^2} dK_{\text{put}} + 2 \int_0^\infty \frac{C(T, K)}{K^2} dK_{\text{call}}, \quad (1)$$

the $1/K^2$ weighting placing most of the mass in the *wings*. This is exactly why a var-swap quote is informative: it constrains the total variance, a global moment that a handful of near-the-money option quotes barely touch. A smile can fit every quoted strike and still carry the wrong wing mass; the var-swap quote pins it.

Heuristic

Think of the var-swap as a single, very reliable “wing quote”. Liquid options cluster near the money, where they say little about the tails; the var-swap is a traded instrument whose value is *dominated* by the tails (the $1/K^2$ weight). Adding it to the objective is the cheapest way to discipline the extrapolation the option quotes leave free — complementary to the Lee wing control of Note 09, which constrains the wing *slope* while the var-swap constrains the wing *mass*.

Invariant

1. One var-swap quote per node, model-independent, shared by the parametric and local-vol views, with its own undo/redo history.
2. With no quote set (or the toggle off) every calibrator is byte-identical to the un-targeted fit.
3. The model’s own fair var-swap is computed by a route that is cheap *inside* the optimizer loop — never by replication against a curve that must be inverted per point.
4. The weight is relative (a fraction of the node’s option weight), so the target means the same thing on a 10-quote node and a 100-quote node.

2 The replication and the penalty

In forward-normalized units ($F = 1$, $k = \log K$), (1) becomes a single integral over the normalized Black call $B(k, w(k))$,

$$w_{\text{vs}} = 2 \int_{-\infty}^{\infty} \left[B(k, w(k)) + (e^k - 1)^- \right] e^{-k} dk, \quad (2)$$

evaluated by a trapezoid on $k \in [-H, H]$ with $H = 6.0$ and 801 points (the put leg adds $(e^k - 1)^-$ for $k < 0$). In code, (2) is a trapezoid over any model’s $w(k)$ (matches production to 10^{-9}):

Listing 1: Model-free var-swap total variance by log-contract replication (2) (distilled from `calib/varswap.py`).

```
1 def varswap_total_variance(implied_w, half=6.0, n=801):    # implied_w(k)
   k) -> total variance
2     k = np.linspace(-half, half, n)
3     w = np.maximum(implied_w(k), 1e-12)
```

```

4     integ = black_call(k, w)*np.exp(-k)                # OTM call leg,
      discounted
5     put = k < 0
6     integ[put] += 1 - np.exp(-k[put])                  # (e^k - 1) e^{-k}
      on the put side
7     return 2.0*np.trapezoid(integ, k)

```

The half-width is coarser than the diagnostics grid because this runs *inside* the optimizer loop; ± 6 captures the OTM mass to well under a basis point for any sane smile, as fig. 1 confirms — the fair var-swap vol has plateaued by a half-width of ~ 4 .

The penalty is a single residual in *volatility* units, so it stacks directly onto the vol-space data residuals of SVI/SIV and the vega-normalized residuals of LQD/LV:

$$r_{vs} = \sqrt{\omega_{vs}} (\sigma_{vs}^{\text{model}} - \sigma_{vs}^{\text{quote}}), \quad \sigma_{vs} = \sqrt{w_{vs}/T}. \quad (3)$$

The weight ω_{vs} is set to a chosen fraction (`varSwapWeightPct`, default 10%) of the node’s summed option-quote weight, so the var-swap contributes a controlled share of the total objective rather than an absolute, scale-dependent amount.

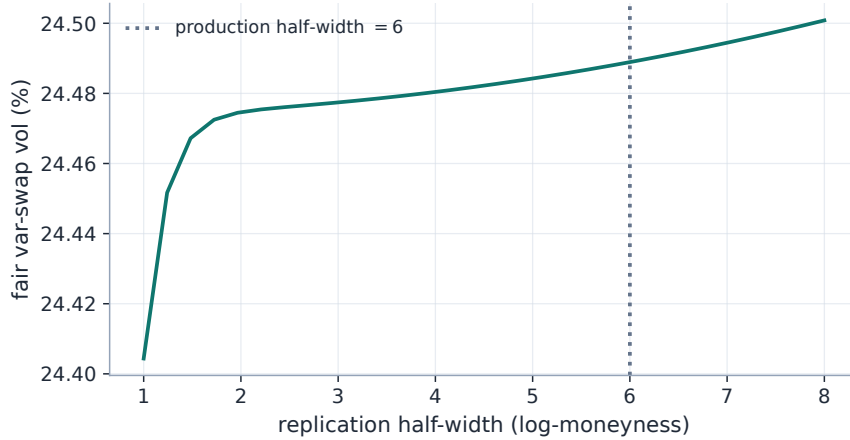


Figure 1: Fair var-swap volatility vs the replication half-width. The integral has converged well before the production half-width of 6.

3 Computing the model’s var-swap cheaply

The penalty needs the *model’s* fair var-swap at every optimizer iteration.

Case file: the fit that took minutes

Setup. The first wiring of the var-swap penalty used the generic replication (2) for every model — correct, uniform, and obviously fine, since the integral is just a trapezoid.

Failure mode. LQD fits with a var-swap target went from milliseconds to *minutes*, hanging the calibrate loop.

Diagnosis. The replication calls the model’s `implied_w(k)` on ~ 800 points — and for LQD each point is a Black *inversion* of a quadrature-priced call. Inside a finite-difference Jacobian

that is ~ 800 root-finds *per parameter column per iteration*: thousands of inversions to price one number the model knows in closed form.

Fix. Per-model routes (below): LQD prices the log contract exactly as $-2\mathbb{E}[X]$ on its existing grid; SVI/SIV keep the replication because their $w(k)$ is arithmetic (evaluated, never solved); the LV surface gets a source PDE with analytic sensitivities.

Verdict. The closed form agrees with the replication to 1.53 vol bps (validating both), and the var-swap penalty now costs essentially nothing in any model. The lesson generalizes: *a uniform implementation is not a virtue when the models' native objects differ — route each model through what it already computes.*

3.1 LQD: the exact closed form

For LQD the var-swap is a one-line consequence of the log-contract identity $\mathbb{E}[-2X] = -2 \int_0^1 Q(u) du$. In the logit coordinate this is a single quadrature against the *same* grid the slice already built,

$$w_{\text{vs}}^{\text{LQD}} = -2 \mathbb{E}[X] = -2 \int_{-\infty}^{\infty} Q(z) \Lambda(z) (1 - \Lambda(z)) dz, \quad (4)$$

whose integrand decays like $z e^{-|z|}$. No inversion, no extra grid — it is essentially free. Table 1 confirms it agrees with the generic replication to 1.53 vol bps, validating both.

3.2 SVI and SIV: arithmetic curves

For SVI and Multi-Core SIV the total variance $w(k)$ is closed-form arithmetic (a hyperbola, or a log-cosh sum), so the replication (2) is cheap directly — no inversion is needed because $w(k)$ is evaluated, not solved.

3.3 Local volatility: the source PDE

The piecewise-affine surface has no arithmetic $w(k)$; pricing each k means a PDE solve. The *source-PDE* method (Note 04) instead reads the var-swap off one backward parabolic solve,

$$\partial_t g + \frac{1}{2} \nu(t, x) x^2 \partial_{xx} g + \nu = 0, \quad g(T, \cdot) = 0, \quad I(T) = g(0, 1), \quad (5)$$

with analytic sensitivities $\partial I / \partial \theta$ from the same multi-RHS machinery as the value solve, and — unlike the replication — no dependence on the truncation x_{max} . It is gated by `varSwapMethod` (default `static` replication; the source PDE is the faster alternative).

4 Worked example

Figure 2 fits an LQD slice to an index-like smile whose *natural* fair var-swap is 24.47%, then refits it with a var-swap quote of 25.97% (1.5 vol points higher) at 10%-class weight. The targeted fit lifts its fair var-swap to 25.91% — almost all the way to the quote — while still threading the option quotes: the var-swap adds wing mass the near-the-money options did not constrain. Table 1 collects the numbers.

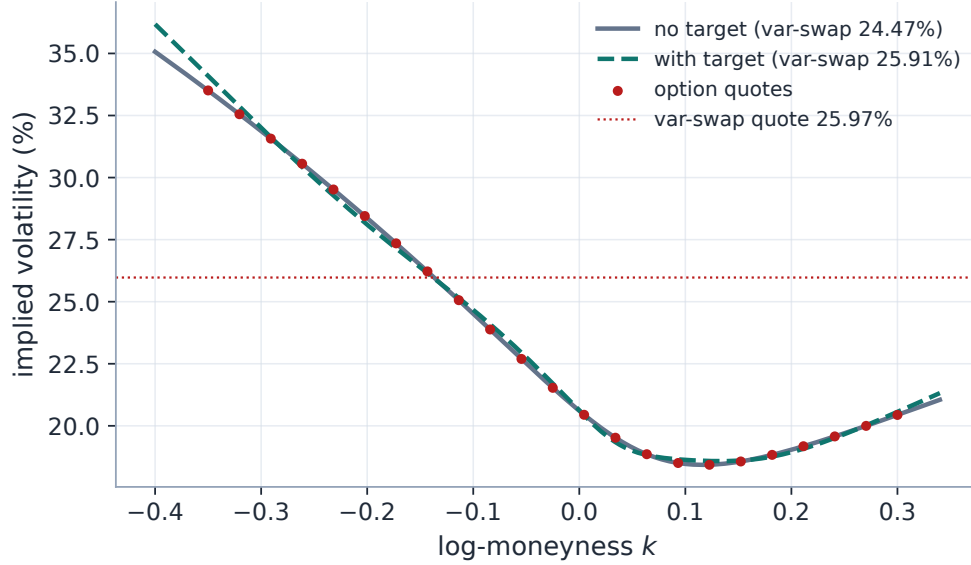


Figure 2: An LQD fit with (dashed) and without (solid) a var-swap target above the smile’s natural level. The target lifts the fair var-swap toward the quote by adding wing mass while keeping the option fit.

Table 1: Var-swap numbers (fresh run): closed form vs replication, and the targeted pull.

Quantity	Value
LQD closed-form var-swap vol	24.47%
Replication var-swap vol	24.49%
Closed-form vs replication agreement	1.53 vol bp
Var-swap quote (target)	25.97%
Targeted-fit fair var-swap	25.91%

5 What is original here

The log-contract replication is Carr–Madan; the contributions are practical. The *per-model cheap var-swap* — LQD’s closed form (4), SVI/SIV’s arithmetic curves, and the LV source PDE (5) with analytic sensitivities — is what makes a var-swap target affordable *inside* the optimizer loop rather than as a slow post-hoc check; and the *relative* weighting (a fraction of the node’s option weight) keeps the target scale-free across nodes with very different quote counts. Together they make one extra global constraint cost essentially nothing.

6 Traceability

Table 2: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
Penalty pulls every model toward the quote (3)		calib/varswap.py	test_varswap.py:: test_penalty_pulls_model_ varswap_toward_quote_all_ models
No target / disabled is identical	byte-invariant 2	calib/varswap.py	test_varswap.py:: test_none_target_ is_byte_identical, ::test_disabling_varswap_ drops_the_penalty
Weight strength monotone (3)		calib/varswap.py	test_varswap.py::test_ weight_strength_monotone
Session set/exclude/undo/redo	invariant 1	api/varswap_session.py	test_varswap.py:: test_session_set_ exclude_include_remove, ::test_session_undo_redo_ and_validation
LV fit carries and honours the quote	section 3	api/affine_fit.py	test_varswap.py:: test_affine_fit_carries_ and_honours_varswap
Source PDE matches static; analytic sensitivities (5)		models/localvol/ varswap_pde.py	test_varswap_source. py::test_source_pde_ value_matches_static, ::test_source_pde_theta_ sensitivity_matches_fd
LQD closed form (the case-file fix) (4)		models/lqd/quadrature.py::var_swap_strike	test_varswap.py (all-models pull), agreement figure 1.53 bp

A Hyperparameter atlas

Table 3: Variance-swap hyperparameters.

Knob	Default	Role
<i>Surfaced</i>		
varSwapEnabled	true	Master toggle (surfaces the var-swap level); the penalty itself activates only when a quote is set on the node.
varSwapWeightPct	10%	Var-swap weight as a fraction of node option weight.
varSwapMethod (LV)	static	Replication or source-PDE pricer.
<i>Hidden</i>		
VS_HALF_WIDTH H	6.0	Replication half-width in log-moneyness.
VS_POINTS	801	Replication trapezoid points.
_W_FLOOR	10^{-12}	Total-variance floor in the integrand.

Performance. The LQD closed form is one quadrature on the existing grid (free); the replication is ~ 800 arithmetic evaluations for SVI/SIV (negligible); the LV source PDE is one extra backward solve with analytic sensitivities. Using the generic replication on an LQD curve — ~ 800 Black inversions per Jacobian column — would turn a millisecond fit into minutes, which is the whole reason for the per-model routes of section 3.

B Reference implementation: the LQD closed form

For LQD the fair var-swap total variance is $-2 \mathbb{E}[X]$ with no replication integral and no Black inversions: the log contract is exactly the mean log-return, and the quadrature grid already carries $Q(z)$ and $u(1-u)$ (Note 01). Agrees with the generic replication to 1.53 vol bps and with production to 10^{-12} .

Listing 2: The LQD closed-form var-swap strike, the mean log-return (distilled from `models/lqd/quadrature.py`).

```

1  def var_swap_strike(z, q_z, u):          # z: logit grid, q_z = Q(z),
    u = expit(z)
2  return -2.0 * np.trapezoid(q_z * u * (1 - u), z)  # = -2 E[X],
    the log contract

```

References

- [1] P. Carr and D. Madan. Towards a theory of volatility trading. In *Volatility*, Risk Books, 1998.
- [2] K. Demeterfi, E. Derman, M. Kamal and J. Zou. A guide to volatility and variance swaps. *J. Derivatives*, 6(4):9–32, 1999.
- [3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.